Data Cleaning and Transformation

Project Documentation

Item	Detail
Author	Hari Hara Kumar
Tool	Microsoft Excel (Excel Table named `Product_DB`)
Dataset	Product Dataset: Product ID, Product Name, Brand Name, Price, Quantity, Category
Records	34 received, 31 after cleaning (3 exact duplicates removed)
Workbook	`Data_Cleaning_and_Transformation.xlsx`

1. Purpose and Approach

Real-world data contains missing values, inconsistent text, duplicate records and poorly structured fields. This project prepares a product dataset for analysis using Excel formulas and formatting tools.

- Every fix is a **formula in a new helper column** (suffix `CL` means cleaned) inside the Excel Table `Product_DB`, so the raw columns stay untouched and each change can be audited.
- Each step below states the problem, the formula, how it works, the result and the supporting screenshot.

2. The Received Data

The snapshot below is the data as received: 34 rows, with blank prices, blank categories, lowercase product names and repeated rows. Narrow columns also cut off some text on screen (for example `Electroni`), which is a display width issue, not a spelling error.

Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	
07-JUN-UK	Headphones	Sony		45	Electroni
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	
08-DEC-DE	Coffee Maker	Nespresso	120	35	
18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni
16-APR-ES	headphones	Bose	250	20	Electroni
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electroni
27-JAN-IT	laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman		10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracke	Xiaomi	150	30	
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
29-SEP-CA	Smartphone	Google	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion
11-JUL-CA	Blender	Vitamix	400	40	Kitchen
16-APR-ES	headphones	Bose	250	20	Electroni
07-MAR-CA	Dress	Zara	60	30	Fashion
13-APR-CA	Toaster	Hamilton	40	10	Kitchen
24-MAY-CA	Fitness Tracke	Garmin	130	5	Electronics
02-DEC-CA	Jeans	Levi's	50	50	Fashion
17-JUN-IN	Laptop	HP	950	25	Electronics
09-JUL-FR	Watch	Casio	100	20	Accessories

Figure 1: Product dataset as received (raw data snapshot)

Data quality audit

Issue	Where	Count	Fix
Exact duplicate rows	Laptop Bag (21-AUG-CA), Headphones Bose (16-APR-ES), Laptop HP (17-JUN-IN)	3	Remove Duplicates
Missing price	Sony Headphones, Coleman Camping Tent, Ray-Ban Sunglasses	3	`IF` + `ISBLANK` + `MODE.SNGL`
Missing category	Backpack, Sneakers (Adidas), Coffee Maker (Nespresso), Fitness Tracker (Xiaomi)	4	Rule-based fill
Inconsistent case	laptop (x2), smartphone, headphones	4 records	`PROPER`
Packed Product ID	e.g. `28-JAN-US` holds day, month and country	all	`LEFT`, `MID`, `RIGHT`
Separate brand and product	e.g. Dell + Laptop	all	`CONCAT`

3. Step 1: Handling Missing Values

3.1 Missing prices

Three products have no price, so totals and averages would silently ignore them. Blanks are filled with the most frequent price in the column (the mode).

```
=IF(ISBLANK([@[Price ($)]]), MODE.SNGL([Price ($)]), [Price ($)])
```

- `ISBLANK` tests whether the price cell is empty.
- If empty, `MODE.SNGL` returns the most common price in the column: **130**.
- If not empty, the original price is kept.

E3 =@IF(ISBLANK([@[Price ($)]]),MODE.SNGL([Price ($)]),[Price ($)])

	A	B	C	D	E	F	G	H
1	Product ID	Product Nam	Brand Nam	Price ($)	Price ($) C	Quantity	Category	
2	28-JAN-US	laptop	Dell	1000	1000	30	Electronics	
3	15-FEB-US	Sneakers	Nike	80	80	15	Fashion	
4	03-MAR-US	Coffee Maker	Keurig	130	130	40	Kitchen	
5	11-APR-US	smartphone	Samsung	900	900	25	Electronics	
6	22-MAY-US	Backpack	North Face	70	70	20		
7	07-JUN-UK	Headphones	Sony		130	45	Electroni	
8	19-JUL-UK	T-shirt	Adidas	30	30	5	Fashion	
9	23-AUG-UK	Blender	Ninja	90	90	35	Kitchen	
10	05-SEP-UK	Tablet	Apple	500	500	50	Electronics	
11	14-OCT-UK	Hiking Boots	Timberland	130	130	10	Outdoor	
12	17-JUN-IN	Laptop	HP	950	950	25	Electronics	
13	25-NOV-AU	Sneakers	Adidas	90	90	40		
14	08-DEC-DE	Coffee Maker	Nespresso	120	120	35		
15	18-FEB-CA	Smartwatch	Fitbit	150	150	15	Electroni	
16	16-APR-ES	headphones	Bose	250	250	20	Electroni	
17	21-AUG-CA	Laptop Bag	Samsonite	50	50	35	Accessories	
18	20-AUG-CN	Smartwatch	Huawei	160	160	15	Electroni	
19	27-JAN-IT	laptop	Asus	980	980	10	Electronics	
20	01-MAR-UK	Sunglasses	Oakley	150	150	15	Fashion	
21	14-AUG-US	Camping Tent	Coleman		130	10	Outdoor	

Figure 2: Price ($) CL formula in the formula bar; blank prices now show 130

Product	Raw price	Cleaned price
Sony Headphones	(blank)	130
Coleman Camping Tent	(blank)	130
Ray-Ban Sunglasses	(blank)	130

Why the mode and not the mean? Prices are right-skewed: laptops and phones sit near 1,000 while most items cost 30 to 160. The mean (297.86) is pulled up by a few expensive items and would overstate a typical product. The median is also 130, which supports the choice. **Impact:** the average price moves from 297.86 (28 priced rows) to 281.61 (all 31 rows).

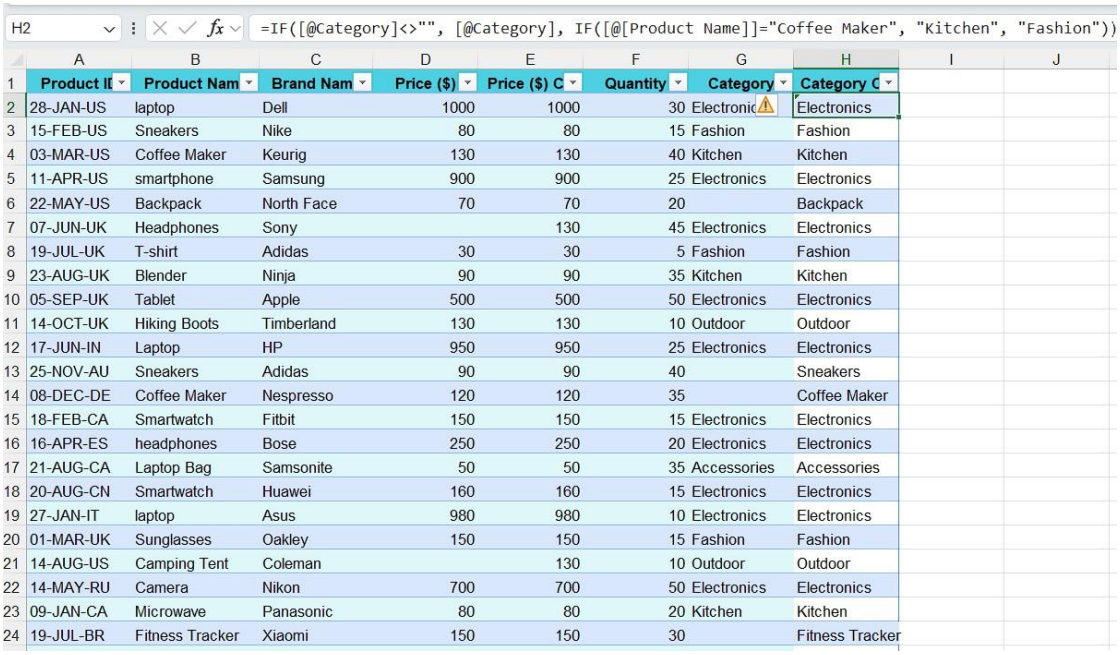
> **Caveat:** prices 130 and 150 each appear three times, so the mode is a tie. `MODE.SNGL` returns the value that appears first in the column (130). Imputed prices are estimates, not facts.

3.2 Missing categories

Four products have no category. Existing categories are kept; blanks are assigned by rule.

```
=IF([@Category]<>"", [@Category], IF([@[Product Name]]="Coffee Maker", "Kitchen",
"Fashion"))
```

- If a category exists, keep it.
- If blank and the product is a Coffee Maker, assign **Kitchen** (the other coffee maker in the data is Kitchen).
- Any other blank becomes **Fashion**.



	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Nam	Brand Nam	Price ($)	Price ($) C	Quantity	Category	Category C		
2	28-JAN-US	laptop	Dell	1000	1000	30	Electronic	Electronics		
3	15-FEB-US	Sneakers	Nike	80	80	15	Fashion	Fashion		
4	03-MAR-US	Coffee Maker	Keurig	130	130	40	Kitchen	Kitchen		
5	11-APR-US	smartphone	Samsung	900	900	25	Electronics	Electronics		
6	22-MAY-US	Backpack	North Face	70	70	20		Backpack		
7	07-JUN-UK	Headphones	Sony		130	45	Electronics	Electronics		
8	19-JUL-UK	T-shirt	Adidas	30	30	5	Fashion	Fashion		
9	23-AUG-UK	Blender	Ninja	90	90	35	Kitchen	Kitchen		
10	05-SEP-UK	Tablet	Apple	500	500	50	Electronics	Electronics		
11	14-OCT-UK	Hiking Boots	Timberland	130	130	10	Outdoor	Outdoor		
12	17-JUN-IN	Laptop	HP	950	950	25	Electronics	Electronics		
13	25-NOV-AU	Sneakers	Adidas	90	90	40		Sneakers		
14	08-DEC-DE	Coffee Maker	Nespresso	120	120	35		Coffee Maker		
15	18-FEB-CA	Smartwatch	Fitbit	150	150	15	Electronics	Electronics		
16	16-APR-ES	headphones	Bose	250	250	20	Electronics	Electronics		
17	21-AUG-CA	Laptop Bag	Samsonite	50	50	35	Accessories	Accessories		
18	20-AUG-CN	Smartwatch	Huawei	160	160	15	Electronics	Electronics		
19	27-JAN-IT	laptop	Asus	980	980	10	Electronics	Electronics		
20	01-MAR-UK	Sunglasses	Oakley	150	150	15	Fashion	Fashion		
21	14-AUG-US	Camping Tent	Coleman		130	10	Outdoor	Outdoor		
22	14-MAY-RU	Camera	Nikon	700	700	50	Electronics	Electronics		
23	09-JAN-CA	Microwave	Panasonic	80	80	20	Kitchen	Kitchen		
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	150	30		Fitness Tracker		

Figure 3: Category CL formula in the formula bar

Product	Raw category	Cleaned category	Method
Backpack (North Face)	(blank)	Fashion	Formula fallback
Sneakers (Adidas)	(blank)	Fashion	Formula fallback (Nike Sneakers is Fashion)
Coffee Maker (Nespresso)	(blank)	Kitchen	Formula rule
Fitness Tracker (Xiaomi)	(blank)	Electronics	Entered manually (other fitness trackers are Electronics)

> The "else Fashion" fallback only works because these blanks happen to be Fashion items. The Fitness Tracker had to be typed by hand for exactly that reason, and Backpack could arguably be Outdoor or Accessories. See Section 10 for a sturdier approach.

4. Step 2: Correcting Inconsistent Data

Four product names are lowercase (`laptop` twice, `smartphone`, `headphones`) while the rest are capitalised. Left alone, `laptop` and `Laptop` would count as different products in a PivotTable.

```
=PROPER([@[Product Name]])
```

`PROPER` capitalises the first letter of each word and lowercases the rest, fixing every row with one formula instead of many Find and Replace passes.

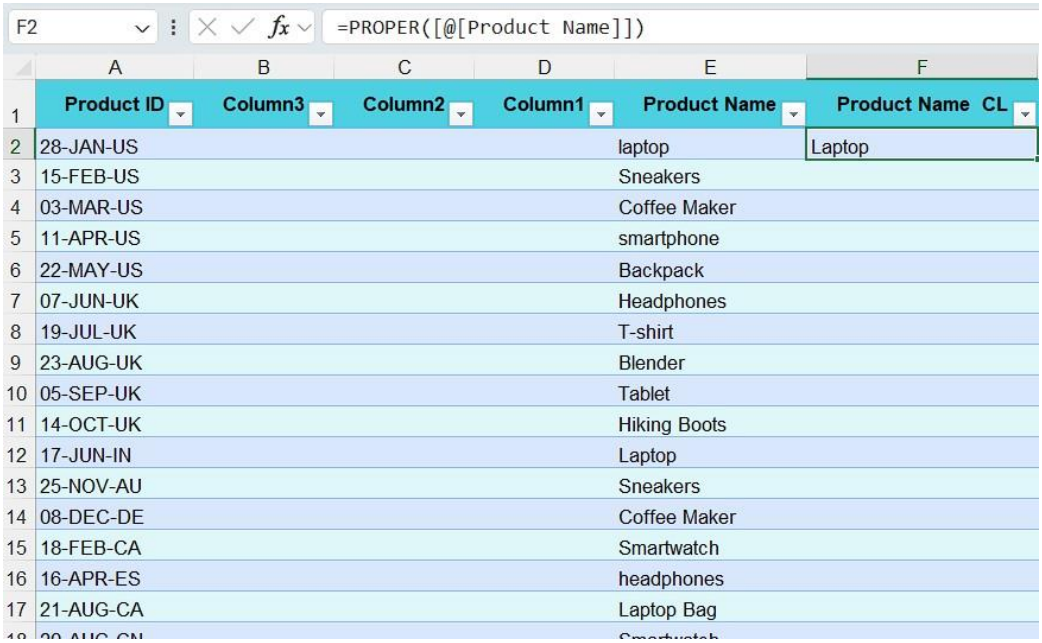


Figure 4: PROPER formula; laptop becomes Laptop

Before	After
Laptop	Laptop
Smartphone	Smartphone
Headphones	Headphones
T-shirt	T-Shirt (PROPER also capitalises after a hyphen)

Category typos: the five category values (Electronics, Fashion, Kitchen, Outdoor, Accessories) are spelled consistently, so no Find and Replace correction was needed. The `Electroni` seen on screen is only column width.

5. Step 3: Removing Duplicates

Data > Remove Duplicates was applied across all columns. Three rows were exact repeats of earlier rows, taking the dataset from **34 to 31 records**.

Product ID	Product	Brand	Price	Qty	Category
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
16-APR-ES	headphones	Bose	250	20	Electronics
17-JUN-IN	Laptop	HP	950	25	Electronics

Removing duplicates first matters: repeated rows would double-count prices in totals, averages and the mode.

> Add a screenshot of the Remove Duplicates dialog and its confirmation message (Excel reports how many duplicates were found and how many unique values remain) as evidence for this step.

6. Step 4: Splitting and Merging Data

6.1 Splitting Product ID

Product IDs follow the pattern `DD-MON-CC` (for example `28-JAN-US`), so each part sits at a fixed position.

New column	Formula	Logic	Example
Product ID No (Day)	`=LEFT([@[Product ID]],2)`	First 2 characters	28
Date Of.Manu (Month)	`=MID([@[Product ID]],4,3)`	3 characters from position 4, skipping the hyphen	JAN
Country Code	`=RIGHT([@[Product ID]],2)`	Last 2 characters	US

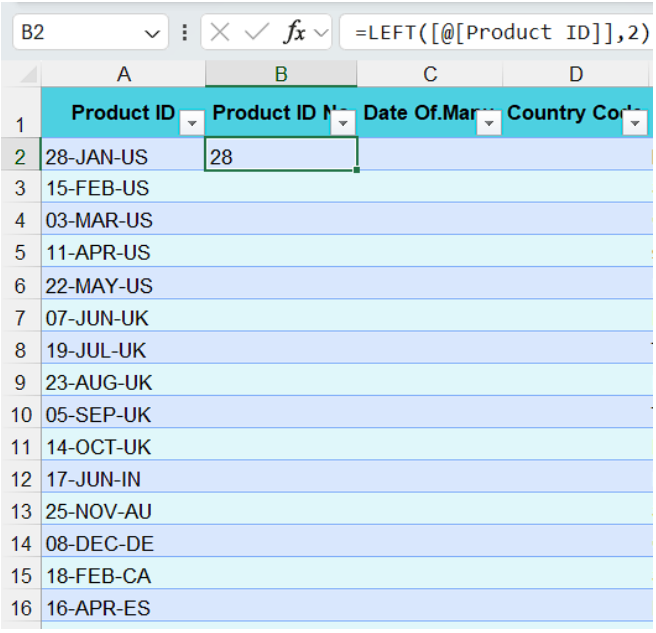


Figure 5: LEFT extracts the day

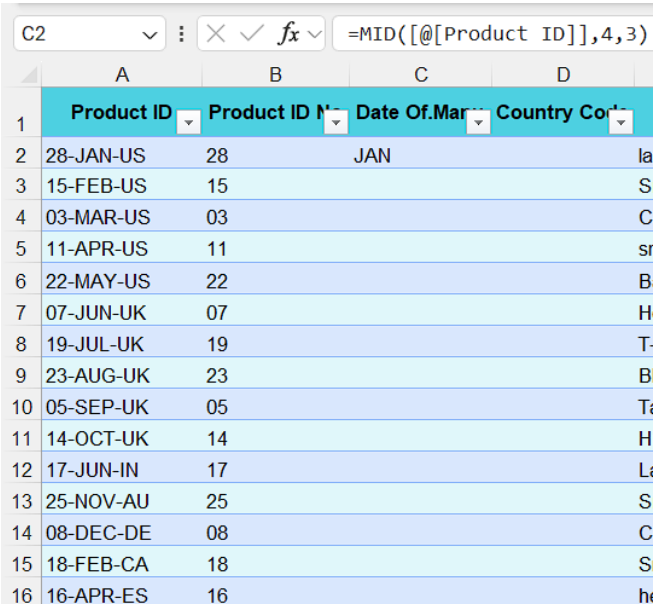


Figure 6: MID extracts the month

D2 =RIGHT([@[Product ID]],2)

	A	B	C	D
1	Product ID	Product ID N	Date Of.Mar	Country Cod
2	28-JAN-US	28	JAN	US
3	15-FEB-US	15	FEB	
4	03-MAR-US	03	MAR	
5	11-APR-US	11	APR	
6	22-MAY-US	22	MAY	
7	07-JUN-UK	07	JUN	
8	19-JUL-UK	19	JUL	
9	23-AUG-UK	23	AUG	
10	05-SEP-UK	05	SEP	
11	14-OCT-UK	14	OCT	
12	17-JUN-IN	17	JUN	
13	25-NOV-AU	25	NOV	
14	08-DEC-DE	08	DEC	
15	18-FEB-CA	18	FEB	
16	16-APR-ES	16	APR	

Figure 7: RIGHT extracts the country code

What the split reveals: 12 country codes across the 31 products. CA is the most common (10 of 31), followed by US and UK (6 each). All 12 months appear.

6.2 Merging brand and product

```
=CONCAT([@[Brand Name]], " ", [@[Product Name CL]])
```

The `" "` adds a space between the parts. Using the cleaned name (`Product Name CL`) keeps the merged label consistently capitalised: `Dell Laptop`, `Samsung Smartphone`, `Bose Headphones`.

H2 =CONCAT([@[Brand Name]]," ",[@[Product Name]])

	A	B	C	D	E	F	G	H
1	Product ID	Column3	Column2	Column1	Product Name	Product Name CL	Brand Name	Brand Prduct Name
2	28-JAN-US				laptop	Laptop	Dell	Dell laptop
3	15-FEB-US				Sneakers	Sneakers	Nike	
4	03-MAR-US				Coffee Maker	Coffee Maker	Keurig	
5	11-APR-US				smartphone	Smartphone	Samsung	
6	22-MAY-US				Backpack	Backpack	North Face	
7	07-JUN-UK				Headphones	Headphones	Sony	
8	19-JUL-UK				T-shirt	T-Shirt	Adidas	
9	23-AUG-UK				Blender	Blender	Ninja	
10	05-SEP-UK				Tablet	Tablet	Apple	
11	14-OCT-UK				Hiking Boots	Hiking Boots	Timberland	
12	17-JUN-IN				Laptop	Laptop	HP	
13	25-NOV-AU				Sneakers	Sneakers	Adidas	
14	08-DEC-DE				Coffee Maker	Coffee Maker	Nespresso	
15	18-FEB-CA				Smartwatch	Smartwatch	Fitbit	
16	16-APR-ES				headphones	Headphones	Bose	
17	21-AUG-CA				Laptop Bag	Laptop Bag	Samsonite	
18	20-AUG-CN				Smartwatch	Smartwatch	Huawei	
19	27-JAN-IT				laptop	Laptop	Asus	
20	01-MAR-UK				Sunglasses	Sunglasses	Oakley	
21	14-AUG-US				Camping Tent	Camping Tent	Coleman	
22	14-MAY-RU				Camera	Camera	Nikon	
23	09-JAN-CA				Microwave	Microwave	Panasonic	

Figure 8: CONCAT merges brand and product name

> This screenshot refers to the raw `Product Name` column, so the result shows "Dell laptop". The final workbook uses `Product Name CL` and gives "Dell Laptop". Retake the screenshot with the final formula so the document matches the workbook.

7. Step 5: Number Formatting

- **Price:** `Price ($) CL` is formatted as currency with two decimals.
- **Date:** the month column holds a three-letter text value (`JAN`) because the Product ID contains no year, so a `DD-MM-YYYY` display format has nothing to act on. Section 10 shows how to build a true date.

8. Step 6: Conditional Formatting

Column	Rule	Purpose
`Price ($) CL`	Data bars (minimum to maximum)	Shows relative price size at a glance
`Category CL`	Custom rule: text contains "Electronics", pink fill	Highlights all 13 Electronics products

Product ID	Product ID M	Date Of.Mar	Country Cor	Product Name	Product Name CL	Brand Name	Brand Prduct Name	Price ($)	Price ($) CL	Quantity	Category	Category CL
28-JAN-US	28	JAN	US	laptop	Laptop	Dell	Dell Laptop	1000	₹ 1,000.00	30	Electronics	Electronics
15-FEB-US	15	FEB	US	Sneakers	Sneakers	Nike	Nike Sneakers	80	₹ 80.00	15	Fashion	Fashion
03-MAR-US	03	MAR	US	Coffee Maker	Coffee Maker	Keurig	Keurig Coffee Maker	130	₹ 130.00	40	Kitchen	Kitchen
11-APR-US	11	APR	US	smartphone	Smartphone	Samsung	Samsung Smartphone	900	₹ 900.00	25	Electronics	Electronics
22-MAY-US	22	MAY	US	Backpack	Backpack	North Face	North Face Backpack	70	₹ 70.00	20		Fashion
07-JUN-UK	07	JUN	UK	Headphones	Headphones	Sony	Sony Headphones		₹ 130.00	45	Electronics	Electronics
19-JUL-UK	19	JUL	UK	T-shirt	T-Shirt	Adidas	Adidas T-Shirt	30	₹ 30.00	5	Fashion	Fashion
23-AUG-UK	23	AUG	UK	Blender	Blender	Ninja	Ninja Blender	90	₹ 90.00	35	Kitchen	Kitchen
05-SEP-UK	05	SEP	UK	Tablet	Tablet	Apple	Apple Tablet	500	₹ 500.00	50	Electronics	Electronics
14-OCT-UK	14	OCT	UK	Hiking Boots	Hiking Boots	Timberland	Timberland Hiking Boots	130	₹ 130.00	10	Outdoor	Outdoor
17-JUN-IN	17	JUN	IN	Laptop	Laptop	HP	HP Laptop	950	₹ 950.00	25	Electronics	Electronics
25-NOV-AU	25	NOV	AU	Sneakers	Sneakers	Adidas	Adidas Sneakers	90	₹ 90.00	40		Fashion
08-DEC-DE	08	DEC	DE	Coffee Maker	Coffee Maker	Nespresso	Nespresso Coffee Maker	120	₹ 120.00	35		Kitchen
18-FEB-CA	18	FEB	CA	Smartwatch	Smartwatch	Fitbit	Fitbit Smartwatch	150	₹ 150.00	15	Electronics	Electronics
16-APR-ES	16	APR	ES	headphones	Headphones	Bose	Bose Headphones	250	₹ 250.00	20	Electronics	Electronics
21-AUG-CA	21	AUG	CA	Laptop Bag	Laptop Bag	Samsonite	Samsonite Laptop Bag	50	₹ 50.00	35	Accessories	Accessories
20-AUG-CN	20	AUG	CN	Smartwatch	Smartwatch	Huawei	Huawei Smartwatch	160	₹ 160.00	15	Electronics	Electronics
27-JAN-IT	27	JAN	IT	laptop	Laptop	Asus	Asus Laptop	980	₹ 980.00	10	Electronics	Electronics
01-MAR-UK	01	MAR	UK	Sunglasses	Sunglasses	Oakley	Oakley Sunglasses	150	₹ 150.00	15	Fashion	Fashion

Figure 9: Cleaned dataset with data bars on Price CL and Electronics highlighted

9. Before vs After

Measure	Before	After
Records	34	31
Exact duplicate rows	3	0
Missing prices	3	0
Missing categories	4	0
Lowercase product names	4	0
Average price	297.86 (28 priced rows)	281.61 (all 31 rows)
Category counts	4 blanks	Electronics 13, Fashion 8, Kitchen 6, Outdoor 2, Accessories 2
Product ID	1 packed text column	Day, Month, Country Code
Brand + product	2 columns	1 merged label

10. Notes and Recommended Refinements

Area	Observation	Suggested improvement
Date format	Month is text, and the Product ID has no year, so `DD-MM-YYYY` cannot display.	`=DATEVALUE(LEFT([@[Product ID]],6))`, then format as `dd-mm-yyyy`. Excel assumes the current year; state that assumption.
Currency symbol	Header says `($)` but the cell format shows the rupee symbol.	Use `$` (English, US) in Format Cells, or rename the header.

Area	Observation	Suggested improvement
Category fallback	"Else Fashion" is a catch-all and needed one manual override.	Keep a Product to Category lookup table and use `XLOOKUP`.
Column names	`Brand Prduct Name` is misspelled; the brief names Product Brand, Manufacturing Date and Day.	Rename the headers to match the brief.
Imputed values	Filled prices and categories are estimates.	Add an `Is Imputed` flag column.
Screenshots to retake	Figure 3 shows blank-category rows returning the product name (Backpack, Sneakers) rather than the final result. Figure 8 uses the raw product name. Some figures show placeholder headers (Column1 to Column3).	Retake with the final formulas filled down the whole column and final headers.

11. Deliverables Checklist

Required item	Where it is covered
Missing value handling	Section 3, Figures 2 and 3
Find and Replace / text corrections	Section 4, Figure 4
Duplicate removal process	Section 5 (add dialog screenshot)
Column splitting and merging	Section 6, Figures 5 to 8
Number formatting	Section 7
Conditional formatting results	Section 8, Figure 9
Brief explanation for each step	Each section